The Placenta Mediates Fetal Dose

Maternal exposure is not equivalent to fetal dose; the placenta transports, metabolizes, restricts, or facilitates substances, including via an incomplete glucocorticoid buffer.

ADE: MODERATE-TO-HIGH-MECHANISM; LOW-CHILD-FORECAST | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFI

WHAT WE KNOW (Narrowest Supported Conclusion)

The placenta is a transient organ of exchange, metabolism, and endocrine signalling. Maternal blood contents are not a fetal dose. Placental 11β-HSD2 inactivates much maternal cortisol; the buffer is real and incomplete.

EVIDENCE AND CAUSAL ASSESSMENT

Burton & Fowden (2015) review transport, uterine-artery remodelling, endocrine modulation of maternal metabolism, and the placenta as a selective barrier. Seckl & Holmes (2007) review 11β-HSD2 as a fetoplacental glucocorticoid barrier: animal knockout/inhibition and human HSD11B2 mutations associate with reduced birthweight; this is mechanism plus selected human genetics, not a population stress-to-IQ path (see the stress card).

EFFECT SIZE AND PERSISTENCE

A placental enzyme level or weight is contemporaneous physiology. Lasting impairment does not follow from the existence of a pathway.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (biological placental transport physiology, not an observational population phenotype association).

Placenta sampled at birth; reverse causation; maternal–fetal genotype; over-translation of programming animal models.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean the mother “sends” every feeling or chemical to the child, or that placental markers are a personality test.

WHAT ACTION IS REASONABLY JUSTIFIED

Clinical attention to placental insufficiency and known teratogens is justified. Individual purification or guilt about ordinary maternal states is not licensed by this mechanism.

VERIFIED SOURCES & DOI IDENTIFIERS

- Burton & Fowden (2015), Phil. Trans. R. Soc. B
- Seckl & Holmes (2007), Nat. Clin. Pract. Endocrinol. Metab.